

Fast Adaptive Insecure Monitoring: the monitoring system that should never have been invented, that is fun

jul

2024-07-07

- Intro
- Mort de FAIM?
- Quickstart
 - Requirements
 - Starting the probe
 - Starting the collect of data
- CAVEAT
 - FreeBSD jails
 - Qemu
- State Machine of the system
- Agent Oriented Programming
- Documentation of each scripts
 - ./bin/asci_plot.sh.txt
 - ./bin/basic_plot.sh.txt
 - ./bin/clock.sh.txt
 - ./bin/launch_lurker.sh.txt
 - ./bin/launch_writer.sh.txt
 - ./bin/lurker.sh.txt
 - ./bin/mkhtml.sh.txt
 - ./bin/plot_histo_g.sh.txt
 - ./bin/plot_histo.sh.txt
 - ./bin/plot_rrd2.sh.txt
 - ./bin/test.sh.txt
 - ./bin/writer.sh.txt
 - ./plugin/cpu.txt
 - ./plugin/ibm_acpi_fan.txt
 - ./plugin/ibm_acpi.txt
 - ./plugin/irq.txt
 - ./plugin/open_files.txt
 - ./plugin/processes.txt
 - ./plugin/stat.txt

– ./plugin/tcp.txt

% Fast Adaptive Insecure Monitoring: the monitoring system that should never have been invented, that is fun % jul % 2024-07-07

Intro

This project is an implementation of such a way of thinking distributed system. For sake of education I took the most compact language for the task : *bash*

We are gonne realize on this principle a Fast Adaptative Insecure Monitoring system.

FAIM is designed as a funny experiment of doing a munin clone (doing less) in bash only that is specialized in high speed (~1 seconde / measure) distributed measuring system without a centralized collector.

No broker, no Zmq, no webrtc, no QUIC, no rabbitMQ are used for transport but ... BROADCAST UDP.

Hence, well, this toy is fundamentally insecure and can hardly be ciphered in its current form. But, it enables a category of software that are both educational for doing your own tool AND for deploying an adhoc measuring system.

Read full documentation [here](#)

Mort de FAIM?

It's funny in french, because it means starving.

But actually, I really don't like the insecure part of this, so as a demonstrator I have tried mqtt as a bus and even though it removes the beauty of no broker, it has proven easy and fun to use.

However, I committed a python proof of concept which defeat the concept of *all in bash*. I will wait for curl to fully support mqtt(s).

Right now, mqtt is incompletely supported by curl, but I will integrate it when it will be ready under the nickname FASM (Fast Adaptative Secure Monitoring).

The PoC with all security turned on is here for the pub part, here for the sub part.

NB: I could hack a version with `mosquitto_pub` and `mosquitto_sub`

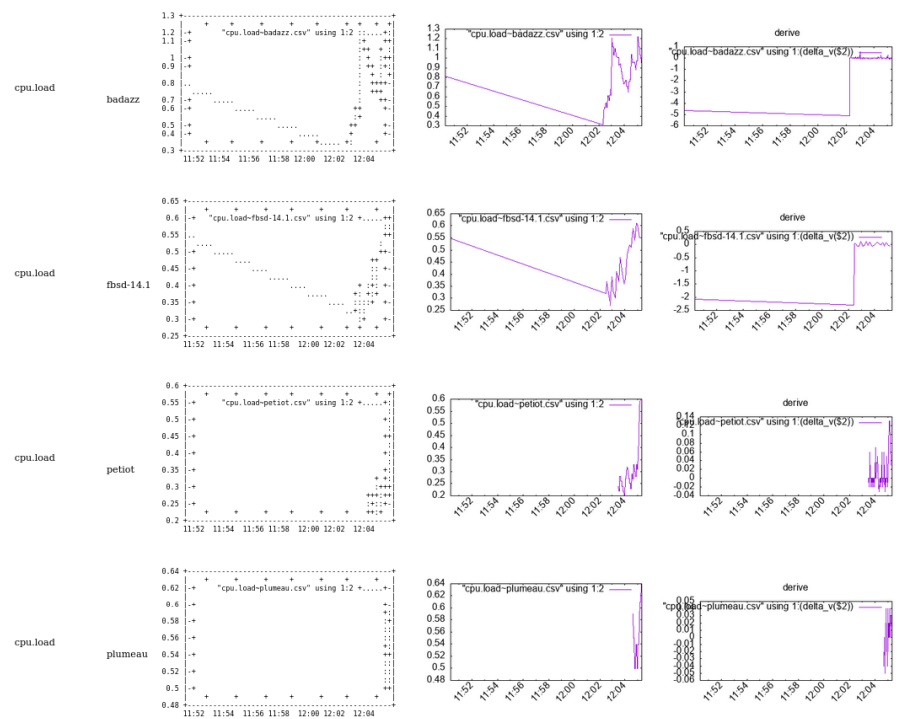


Figure 1: example

Quickstart

Requirements

Perl, python3, gnuplot (gnuplot-lite maybe enough if 1Gb dependency rebukes you), bash, socat, and whatever the plugins have dependencies upon.

Starting the probe

```
./start.sh
```

Starts the probe. It will emit (see API of start for network parameters) on the UDP broadcast address in ASCII exactly what `bin/writer.sh` emits on stdout.

To stop the probe simply type

```
./stop.sh
```

Starting the collect of data

```
LURKER=1 ./start.sh
```

Will start the probe AND the data collector. The data collector can also be caught in a standalone mode with `./bin/launch_lurker.sh` or `./bin/launch_lurker.py`.

if you go in `./data/` you will see both csv where data are stored accumulating and the making of the html resume.

Just open `./data/index.html` to view the graphs.

You can erase the content of data at any moment, everything will reconstruct itself.

What the lurker sees from the broadcast is log into `./log/journal.txt`.

CSV files and html can be reconstructed by typing

```
cat log/journal.log | bin/lurker
bin/mkhtml.sh
```

mkhtml is the bash equivalent of PHP or using jinja in python : dynamic html generation.

I seriously advise to install tcpdump, and remember that `tcpdump -A [-i interface] -s0 udp and port 6666` can be a serious life saviour while troubleshooting.

CAVEAT

FreeBSD jails

In order for freebsd Jails to access UDP broadcast, you need to setup a VNET jails

Qemu

In order for qemu guests to access UDP broadcast, you need to setup a bridge

State Machine of the system

See diag.dot

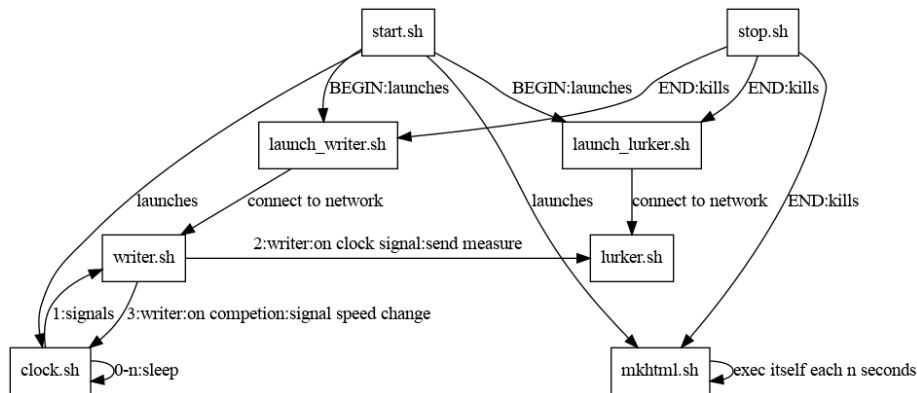


Figure 2: diag

Agent Oriented Programming

1. EverythingIsAnAgent
2. Agent communicate by sending and receiving messages (the way they prefer as long as they send messages).
3. Agent have their own memory and autonomy
4. Every Agent is an instance of an artifact (which then as to be accounted as an agent).
5. The Agent is accountable for maintaining its consistency as a state/transition agent
6. The topology is more important than the code.

7. each state machine is on a plane for which an uncoupled state machine lays.
8. violation of uncoupling between layers is bad so it has to be handled with care.

Documentation of each scripts

API of each components.

./bin/asci_plot.sh.txt

./bin/basic_plot.sh.txt

./bin/clock.sh.txt

./bin/launch_lurker.sh.txt

NAME

launch_lurker.sh

SYNOPSIS Make data collector available for listening to the probes

[TICK=2] [BROADCAST=192.168.1.255] [RANGE=24] [PORT=6666] ./launch_writer.sh

OPTIONS For explanation of options see <file:../start.sh.html>

./bin/launch_writer.sh.txt

NAME

launch_writer.sh

SYNOPSIS Make writer emit on BROADCAST/RANGE on port PORT

[TICK=2] [BROADCAST=192.168.1.255] [RANGE=24] [PORT=6666] ./launch_writer.sh

OPTIONS For explanation of options see <file:../start.sh.html>

./bin/lurker.sh.txt

NAME

lurker.sh

SYNOPSIS Collector of data

`./lurker.sh`

Can be used as

```
while [ 1 ]; do writer.sh | lurker.sh; sleep 30; done
```

To collect data emitted locally about the machine.

Results are written in `../data`

`./bin/mkhtml.sh.txt`

NAME

`mkhtml`

HTML maker

SYNOPSIS Generator of HTML output from data collected in `../data`

```
[DAEMON=] [SINCE=3600] mkhtml.sh
```

Can be used as

```
./mkhtml.sh
```

to generate the web page in `../data`

OPTIONS

DAEMON This code will run permanently waking itself up to update the web page.

SINCE The window span time you are interested in in seconds from NOW

`./bin/plot__histo__g.sh.txt`

`./bin/plot__histo.sh.txt`

`./bin/plot__rrd2.sh.txt`

`./bin/test.sh.txt`

`./bin/writer.sh.txt`

NAME

`writer.sh`

SYNOPSIS Emitter of data

```
[TICK=2] ./writer.sh
```

OPTIONS For explanation of options see "start.sh.html" in .

If TICK is set then writer will assume it is to be launched in conjunction with "clock.sh.html" in . and do nothing until clock.sh sends a signal to it to write data.

./plugin/cpu.txt

NAME

cpu - FAIM plugin to monitor the CPU load

APPLICABLE SYSTEMS

FreeBSD, linux

USAGE

Just create cpu_enabled in the plugin dir

BUGS

None known.

VERSION

v1.1 - 2024-03-24

AUTHOR

Julien Tayon (julien@tayon.net)

LICENSE

GPLv2

./plugin/ibm_acpi_fan.txt

NAME

acpi_ibm - Munin plugin to monitor the fan speed returned by ACPI probe.

APPLICABLE SYSTEMS

FreeBSD systems with ACPI support. man acpi_ibm(4)

CONFIGURATION

add ibm_acpi in loader.conf

USAGE

Link this plugin to @@CONFDIR@@/plugins/ and restart the munin-node.

INTERPRETATION

The plugin shows the fans' speeds.

MAGIC MARKERS

```
### family=auto
### capabilities=autoconf
```

BUGS

None known.

VERSION

v1.1 - 2024-03-24

AUTHOR

Julien Tayon (julien@tayon.net)

LICENSE

GPLv2

./plugin/ibm_acpi.txt

NAME

acpii_ibm - Munin plugin to monitor the temperature in different ACPI Thermal zones.

APPLICABLE SYSTEMS

FreeBSD systems with ACPI support. man acpi_ibm(4)

CONFIGURATION

add ibm_acpi in loader.conf

USAGE

Link this plugin to @@CONFDIR@@/plugins/ and restart the munin-node.

INTERPRETATION

The plugin shows the temperature from the different thermal zones.

MAGIC MARKERS

```
### family=auto
### capabilities=autoconf
```

BUGS

None known.

VERSION

v1.1 - 2024-03-24

AUTHOR

Julien Tayon (julien@tayon.net)

LICENSE

GPLv2

./plugin/irq.txt

NAME

interrupts - list number of interrupts since boot (linux) or the interrupt rate per interrupt

CONFIGURATION

No configuration

AUTHOR

Idea and base from Ragnar Wisløff.

LICENSE

GPLv2

MAGIC MARKERS

```
### family=auto
### capabilities=autoconf
```

POD ERRORS

Hey! The above document had some coding errors, which are explained below:

Around line 16: Non-ASCII character seen before =encoding in 'Wisløff'.
Assuming UTF-8

./plugin/open_files.txt

NAME

open_files - Plugin to monitor the number of open files in the system

CONFIGURATION

No configuration

AUTHOR

Unknown author

LICENSE

GPLv2

MAGIC MARKERS

```
### family=auto
### capabilities=autoconf
```

./plugin/processes.txt

NAME

processes - Plugin to monitor processes and process states.

ABOUT

This plugin requires munin-server version 1.2.5 or 1.3.3 (or higher).

This plugin is backwards compatible with the old processes-plugins found on SunOS, Linux and *BSD (i.e. the history is preserved).

All fields have colours associated with them which reflect the type of process (sleeping/idle = blue, running = green, stopped/zombie/dead = red, etc.)

CONFIGURATION

No configuration for this plugin.

AUTHOR

Copyright (C) 2006 Lars Strand

LICENSE

GNU General Public License, version 2

MAGIC MARKERS

```
### family=auto
### capabilities=autoconf
```

./plugin/stat.txt

NAME

interrupts - Plugin to monitor the number of interrupts and context switches on a system.

CONFIGURATION

No configuration

AUTHOR

Idea and base from Ragnar Wisløff.

LICENSE

GPLv2

MAGIC MARKERS

```
### family=auto
### capabilities=autoconf
```

POD ERRORS

Hey! The above document had some coding errors, which are explained below:

Around line 17: Non-ASCII character seen before =encoding in 'Wisløff.'. Assuming UTF-8

./plugin/tcp.txt

NAME

tcp - Plugin to monitor IPV4/6 TCP socket status on a Linux host.

LICENSE

GPLv2